

Engineering Mathematics

BRANCH: COMMON / BASIC SCIENCES

COURSE CODE: MA101

Subject Overview & Scope

This comprehensive document serves as the official compiled study notes for Engineering Mathematics (MA101) under the Common / Basic Sciences department. It is structured into four core university syllabus units covering primary concepts, definitions, formulas, and advanced technical applications for university examinations and placements.

University Course Syllabus & Core Units

Unit 1: Multivariable Calculus

Limits, continuity, partial derivatives, chain rule. Maxima, minima of two variables, Lagrange multipliers. Double and triple integrals.

Unit 2: Ordinary Differential Equations

First-order linear ODEs, exact equations, Bernoulli's equation. Higher-order linear ODEs with constant coefficients. Method of variation of parameters.

Unit 3: Linear Algebra

Vector spaces, linear independence, basis, dimension. Matrices, rank, system of linear equations. Eigenvalues, eigenvectors, Cayley-Hamilton theorem.

Unit 4: Probability & Statistics

Conditional probability, Bayes' theorem. Random variables, binomial, Poisson, normal distributions. Correlation and regression.

Bayes Theorem: $P(A|B) = P(B|A) \cdot P(A) / P(B)$. Eigenvalue equation: $\det(A - \lambda I) = 0$.

Key Formulas & Exam Pointers

CRITICAL FORMULA / PRINCIPLE:

